Semi-Supervised Labels and Outcome Prediction for Clinical Trials

CS5785 / ORIE5750 / ECE5414 — Midterm Project

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Abstract

Part 1 asks why a trial stopped early. Only 157 of 1,171 training trials have visible reason labels. We keep the 150 reason embeddings and discard eligibility text, diseases, drugs, and start year, because those inputs did not raise the validation score. A Gaussian mixture only helps decide which unlabeled trials are safe to learn from. Logistic regression, cosine neighbors, and shrinkage discriminant analysis then name the reasons. The reported Kaggle sample-wise F1 is 0.89636.

Part 2 asks whether a trial meets its primary endpoint. The score is log loss, and the test trials start later than the training trials. We upweight recent trials, average in a neighbor model that also prefers recent matches, and mix in 30% of an earlier text-and-tree ensemble that errs on different trials. The reported Kaggle log loss is 0.58465. A blend that looked better on our historical checks scored 0.58492, so we kept the first result.

Team responsibilities. All three authors worked on both parts and on this report. **Michael Mazzola** led the Part 1 model. **Tarun Mandalapu** built the first Part 2 classifier, and **Anirudha Hadagali** led the later Part 2 tuning. Tarun Mandalapu drafted the longer project notes that this summary condenses.

1. Part 1: stopping reasons

1.1. Data and the score

The files hold 1,561 trials that stopped early [2, 8]. Training has 1,171 trials: 157 with real labels and 1,014 with every reason marked -100, which means unknown. Validation has 78 labeled trials. Test has 312 later trials, from 2010 to 2016. A trial can carry more than one of Safety, Funding, Efficacy, Recruitment, and Other. Among the 235 real labels, Recruitment appears on 64% and Funding on 5%. The labeled trials are mostly early, so those rates will not match the test set.

The selection score is sample-wise F1, the average of a per-trial F1. We also report F1 for each reason and the average of those five scores. One validation trial moves sample-wise F1 by about 1.3 points, so we rechecked every choice with eight repeats of five grouped folds, once on the 157 training labels and once on all 235, plus three time-ordered windows.

1.2. What the model does

Each trial is represented by 150 numbers from its stopping-reason text. We standardize them using only rows the current split may learn from. IDs are used to line up rows and are not features. Adding start year, eligibility text, disease names, pathways, or drug fingerprints [3, 4, 6] never raised validation sample-wise F1 on the earlier self-training model (0.893 for embeddings alone, 0.891 with year, 0.876 with eligibility or disease text, 0.889 with fingerprints). The embeddings already summarize the text the labels were written from. The other fields describe the trial, and year points the wrong way because the labels are early and the test is late.

Missing labels have to come from an unsupervised step. k-means and DBSCAN were useful checks, and one DBSCAN-plus-logistic setup reached about 0.910 validation F1 under an earlier protocol. We did not submit it. Clustering mostly separates Recruitment from everything else. The piece we kept is a 30-component diagonal Gaussian mixture on 30 principal components, about 77% of the variance [1, 5]. Components start from k-means++, three restarts, `random_state` $= 0$. Expectation–maximization stops when the average log-likelihood improves by less than 10^{-3}, or after 100 iterations. The kept run converged in 29. Each component becomes a reason profile by averaging the real labels assigned to it, plus one pseudo-count of the overall rates. In the teaching step this profile has a 5% vote. The classifier has the other 95%.

The classifier is one logistic regression over the five reasons ($C = 0.1$), not five separate classifiers. A two-reason trial is entered once per reason, with its weight split, so it is not counted twice. Class weights come from the real

labels only and then stay fixed. A generated row counts as a quarter of a real row. We average three scores with fixed weights: logistic regression 50%, cosine 10-nearest neighbors 30%, and shrinkage linear discriminant analysis 20%. Neighbors vote with squared cosine similarity, and a neighbor that has two reasons votes for both. Only the mixture uses the 30 components. The classifiers see all 150 standardized embeddings.

Unlabeled trials are added in two rounds [7, 10]. A trial is accepted when its best reason scores at least 0.8. Each round starts again from the real labels. On the final fit the rounds accept 685 and then 757 trials, with total weight about 103 and then 120, against 235 for the real labels. We always output the top reason, plus any other reason at 0.3 or above, because an empty prediction is always wrong. Test rows are used only for the final predictions. The 0.891 validation score belongs to a fit on 157 labels. The submitted file is retrained on all 235, so that validation number is not an independent score of the submitted file. An earlier 0.89423 submission had fit its scaler and mixture with the test rows included. We stopped doing that.

2. Part 2: trial outcomes

2.1. Data and the score

Each row is one trial in one phase. A combined-phase trial appears twice and those rows stay together in every split. Training has 8,037 rows through 2013, validation has 1,013 rows from the same years, and the test file has 1,500 later rows, 500 per phase, from 2013 to 2022. Kaggle averages the three phase log losses. We select on that same average. F1 at 0.5, ROC-AUC, and precision-recall AUC are diagnostics. Part 1 reasons are joined only after prediction, for the error table. They are not model inputs.

2.2. What the model does

The baseline is logistic regression on eligibility text: word and bigram counts, at most 35,000 terms, minimum document frequency 3, sublinear counts, three phase indicators, and $C = 2$. Separate phase-only fits overfit Phase I. Drug fingerprints did not reliably help, and a matched drug is less common in test (39%) than in training (46%). Naive Bayes ranked trials but was far too confident for log loss.

The model we built on is a stack of eight classifiers [9]: logistic models on eligibility text, diseases, drug names, codes, drug annotations, and metadata, plus naive Bayes and a 50-neighbor model on the eligibility text. A second logistic regression learns the weights from out-of-fold predictions, five folds, with a trial never split across folds. Recent trials get more weight, with a three-year exponential time constant, so a trial loses about half its weight every two years. That stack, refit on all 9,050 labeled rows, scored 0.58688. The same stack without recency weights scored 0.58810. Our first ensembles had used only the 8,037 training rows, so part of the earlier gap was the extra data.

Model	Train	Pooled	Val.	Time	Kaggle
Final blend	**.9321**	.9214	.891	**.884**	**.89636**
LDA ensemble	.9318	**.9216**	**.906**	.878	.89401
Earlier LDA blend	.9281	.9154	.872	.874	—
First submission	.9196	.9098	.891	.861	.89423

Table 1. Part 1 on shared grouped checks. Train uses 157 labels, pooled uses 235, validation is the original 78 trials, and time pools three chronological windows. Higher is better.

The neighbor piece inside the stack ignored time and pulled estimates toward a 50% success rate. File 10 does not replace the stack. It averages it, half and half, with a second stack that decays faster (two-year time constant) and that also sees eight neighborhood estimates: several sizes, sharper similarity, and similarity that includes disease, drug, and ICD text. The weights are fixed. This improved every time-ordered window and scored 0.58609, even though the random validation log loss got slightly worse.

File 10 is weaker when the years are mixed. Our earlier ensemble is weaker on later years. That ensemble is half three logistic models, a quarter a depth-6 tree, and a quarter a depth-8 tree. We first added 20% of it (Kaggle 0.58474). Raising that share to 30% gained only another 0.00009, so we stopped. The submitted probability is

$$0.70\,P_{10}+0.30\,P_{\text{early}}, \quad P_{10} = 0.50\,P_{\text{recency}}+0.50\,P_{\text{fast neighbors}}.$$

Supervised pieces are refit on all 9,050 labeled rows after the recipe is frozen. Test labels are never used. In the inherited stack, word lists may see unlabeled test text. That changes the features, not the answers. A later blend with an extra tree and a phase trend looked better locally and scored 0.58492, so we did not replace file 13. One upload slot was left unused.

3. Results

Table 1 is why Part 1 submitted the final blend rather than the linear-discriminant ensemble. The two are tied under cross-validation. The ensemble wins the 78-trial validation score. The final blend wins the time-ordered check and the reported Kaggle score. On those 78 trials its per-reason F1 is 0.941 Safety (9 trials), 0.667 Funding (4), 0.759 Efficacy (14), 0.990 Recruitment (50), and 0.471 Other (10): sample-wise 0.891, macro 0.765.

Table 2 is the Part 2 path. Lower log loss is better. File 13 is the submission. File 15 wins both local columns and loses on Kaggle, which is why the local ranking was not enough.

Table 3 is the supplied validation split, with preprocessing fit only on that split's training rows. The recency stack improves Phase III log loss from 0.529 to 0.516 and Phase I F1

Model	Time	Grouped	Kaggle
Phase-specific ensemble	—	—	0.59562
Text-heavy ensemble	—	—	0.58993
Stack, no recency	0.5421	0.5797	0.58810
Recency stack	0.5374	0.5874	0.58688
File 10, recent neighbors	0.5359	0.5869	0.58609
File 13, 30% early ensemble	0.5334	0.5790	**0.58465**
File 15, tree and trend	**0.5327**	**0.5783**	0.58492

Table 2. Part 2 log loss. Time averages the forward windows. Grouped is trial-grouped cross-validation. The first two submissions used an earlier local protocol, so those cells are blank.

Phase	Metric	Baseline	Recency	File 10
I	F1	0.765	**0.793**	0.762
	ROC-AUC	0.783	**0.794**	0.787
	PR-AUC	**0.835**	0.830	0.824
	log loss	0.556	0.556	0.560
II	F1	**0.657**	0.647	0.638
	ROC-AUC	0.740	0.753	**0.754**
	PR-AUC	0.717	**0.732**	0.730
	log loss	0.600	0.593	0.593
III	F1	**0.833**	0.825	0.829
	ROC-AUC	0.774	**0.790**	0.786
	PR-AUC	0.862	**0.882**	0.877
	log loss	0.529	**0.516**	0.521

Table 3. Part 2 validation by phase. F1 uses a cutoff of 0.5. The other rows use the probability of success. Phase sizes are 149, 501, and 363. Best value per row in bold. Tied log losses are left unbolded.

from 0.765 to 0.793. It does not improve Phase II F1. File 10 is slightly worse than the recency stack on this random split and better on the later-year windows in Table 2. That is why a higher validation F1 was not the submission rule.

Table 4 counts failed validation trials that the outcome model called successful. The ensemble is the early text-and-tree model that later became 30% of file 13. The readable groups are Phase II Recruitment and the three no-reason groups. Phase III failures are called successful much more often than Phase II failures. Most reason tags are Part 1 predictions. The later neighbor model still called all 9 Phase III efficacy-tagged failures successful.

4. Why these choices

The Part 1 mixture is a poor labeler and a useful tie-breaker. On its own it was the weakest model we tried, because extra components split Recruitment rather than isolating Safety or Funding. A supervised model still reaches 0.941 F1 on Safety from the same embeddings, so the features are not the problem. Cutting the mixture's vote to 5% and the generated-row weight to one quarter lets a confident guess help without outvoting the 235 real labels.

Phase	Reason	n	Base	Ens.
I	Safety	9	.56	.44
I	Funding	4	.50	.50
I	Efficacy	2	.50	.50
I	Recruitment	22	.36	.27
I	Other	5	.80	.80
I	No reason	28	.43	.39
II	Safety	15	.40	.40
II	Funding	4	.00	.25
II	Efficacy	10	.30	.30
II	Recruitment	87	.20	.16
II	Other	18	.33	.28
II	No reason	144	.35	.34
III	Safety	6	.50	.67
III	Funding	0	—	—
III	Efficacy	9	.89	.67
III	Recruitment	21	.52	.57
III	Other	8	.63	.75
III	No reason	82	.56	.55

Table 4. Share of failed validation trials predicted successful. "No reason" means the trial is absent from Part 1. A trial with several reasons is listed under each.

Extra Part 1 features fail for a mechanical reason. The embeddings are a compressed copy of the text the labels came from. Criteria, diseases, and drugs describe what the trial was. Adding them means dozens or thousands of new inputs on 157 real labels, and 44% of trials have no matched drug. Start year looks predictive only because the labeled trials were chosen to be early.

We did not follow Part 1's higher validation score. Four Funding trials and ten Other trials can move the macro average by several points. The time-ordered windows look more like the test years, and they favor the submitted blend, as does the reported Kaggle score.

In Part 2, log loss and F1 disagree on purpose. A model that moves an uncertain trial across 0.5 can lose F1 and still assign the better probability. The test file is a later, more successful cohort than the random validation split, so we trusted time-ordered checks. Recency weights and recent neighbors are two ways of saying that an old similar trial is a weaker example. The 30% older ensemble is there because it misses different trials, and the second 10% of that share was only worth 0.00009 on Kaggle. After file 15 lost on the leaderboard, a better historical average was no longer a reason to upload.

The reason table says where outcome errors live, not how to tune a cutoff. Failed Phase III trials often ran to the end and simply missed the endpoint, so the eligibility text looks like a success. Recruitment and funding failures are more administrative, and the ensemble catches a few more of them.

Safety, Funding, and Other are small in every phase. Only Phase II Recruitment and the no-reason groups are large, and the no-reason group is a lookup rather than a prediction.

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